

Multivariate Analysis: CA Week Five

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Question 1

We are constructing confidence intervals for:

Individual t:

$$\bar{x}_i \pm t_{n-1}(\alpha/2) \sqrt{\frac{s_{ii}}{n}}$$

Simultaneous t:

$$\bar{x}_i \pm \sqrt{\frac{p(n-1)}{(n-p)} F_{p,n-p}(\alpha)} \sqrt{\frac{s_{ii}}{n}}$$

Bonferroni:

$$\bar{x}_i \pm t_{n-1}\left(\frac{\alpha}{2p}\right) \sqrt{\frac{s_{ii}}{n}}$$

Table 1: Confidence intervals for MaxBreadth

	Lower Bound	Upper Bound
MB_Individual_t	129.4514	133.2820
MB_Simultaneous_T2	128.0909	134.6425
MB_Bonferroni	128.8727	133.8607

Table 2: Confidence intervals for BasHeight

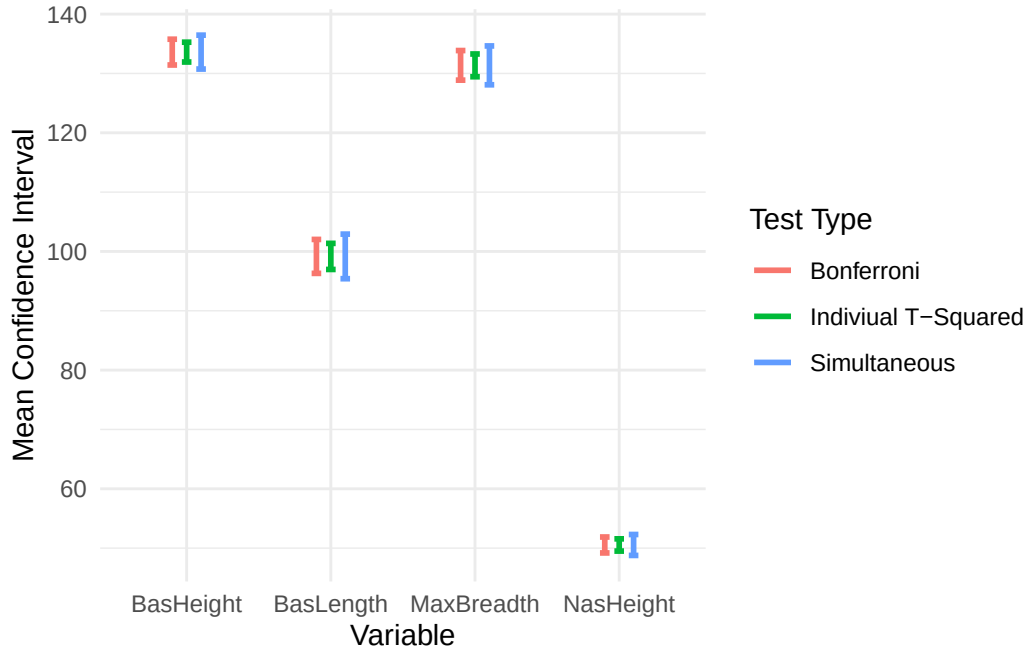
	Lower Bound	Upper Bound
BH_Individual_t	131.9312	135.2688
BH_Simultaneous_T2	130.7458	136.4542
BH_Bonferroni	131.4270	135.7730

Table 3: Confidence intervals for NasHeight

	Lower Bound	Upper Bound
BL_Individual_t	96.96939	101.3639
BL_Simultaneous_T2	95.40858	102.9247
BL_Bonferroni	96.30548	102.0279

Table 4: Confidence intervals for NasHeight

	Lower Bound	Upper Bound
NH_Individual_t	49.50144	51.56523
NH_Simultaneous_T2	48.76844	52.29822
NH_Bonferroni	49.18965	51.87702



As expected the Bonferroni interval is in between the individual t and simultaneous t.

Question 2

we are testing for

$$H_o : \mu_{initial} = \mu_1 = \mu_2 = \dots = \mu_4 \quad vs \quad H_1 : \text{At least one } \mu_j \text{ differs}$$

which is equivalent to:

$$H_o : \mu_{initial} - \mu_1 = \dots = \mu_3 - \mu_4 \quad vs \quad H_1 : \text{At least one } \mu_j \text{ differs}$$

And we define our contrast matrix

$$C = \begin{pmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{pmatrix}$$

Thus, the hypotheses are now,

$$H_o : C\mu = 0 \quad vs \quad H_1 : \text{At least one } \mu_j \text{ differs}$$

Where,

$$C\mu = \begin{pmatrix} \mu_{initial} - \mu_1 \\ \mu_1 - \mu_2 \\ \mu_2 - \mu_3 \\ \mu_3 - \mu_4 \end{pmatrix}$$

Therefore, the resulting simultaneous T^2 confidence interval is

Table 5: Confidence intervals for repeated measures

Lower Bound	Upper Bound
0.399	0.511
1.072	1.793
-0.923	-0.717
-2.053	-0.997

low p-value, some strong evidence for rejecting the null hypothesis. The intervals above do not include 0.